VERIFICATION CERTIFICATE FOR THE LEAN 4 DEVELOPMENT
R3-BIOT-SAVART-STREAM

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ABSTRACT. This document certifies that the Lean 4 formalization in the repository `r3-biot-savart-stream` builds with its pinned toolchain, contains no `sorry` and no user-declared axioms, and that every theorem depends only on Lean's standard axioms.

1. VERIFICATION CERTIFICATE

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/r3-biot-savart-stream`
- Verified commit:
  `00d743e7a586de86e997780628aaa3208b1e8a27`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/r3-biot-savart-stream/actions/runs/41572324283`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/r3-biot-savart-stream/archive/00d743e7a586de86e997780628aaa3208b1e8a27.tar.gz`
  `9130d41bb616cd51fcfdf8015c0d4e01a11193d4b501dad8acde734f4666003b`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 49 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`R3BiotSavart.DivergenceFreeTestField.continuous`	`[propext, Classical.choice, Quot.sound]`
`R3BiotSavart.DivergenceFreeTestField.differentiable`	`[propext, Classical.choice, Quot.sound]`
`R3BiotSavart.DivergenceFreeTestField.integrable`	`[propext, Classical.choice, Quot.sound]`
`R3BiotSavart.biotSavart_add_smul`	`[propext, Classical.choice, Quot.sound]`
`R3BiotSavart.gateauxOn_unique`	`[propext, Classical.choice, Quot.sound]`
`R3BiotSavart.hasDerivAt_kineticEnergy`	`[propext, Classical.choice, Quot.sound]`

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Lean declaration	#print axioms
R3BiotSavart.integral_norm_add_smul_sq	[propext, Classical.choice, Quot.sound]
R3BiotSavart.kineticEnergyCurlIdentityOn	[propext, Classical.choice, Quot.sound]
R3BiotSavart.kineticEnergy_add_smul	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_kineticEnergyCurlIdentityReg_of_divergenceFreeTestField	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_regularStreamRepresentative_of_divergenceFreeTestField	[propext, Classical.choice, Quot.sound]
R3BiotSavart.streamRepresentative	[propext, Classical.choice, Quot.sound]
R3BiotSavart.continuous_cross	[propext, Classical.choice, Quot.sound]
R3BiotSavart.integrable_biotSavart_integrand	[propext, Classical.choice, Quot.sound]
R3BiotSavart.integrable_indicator_ball_rpow_neg_two	[propext, Classical.choice, Quot.sound]
R3BiotSavart.measurable_kernelProfile	[propext, Classical.choice, Quot.sound]
R3BiotSavart.norm_cross_le	[propext, Classical.choice, Quot.sound]
ArnoldGeometric.CoadjointOrbit.property	[propext, Classical.choice, Quot.sound]
ArnoldGeometric.divergenceFree_iff	[propext, Classical.choice, Quot.sound]
ArnoldGeometric.gateaux_representative_unique	[propext, Classical.choice, Quot.sound]
ArnoldGeometric.orbit_add_smul_div_free	[propext, Classical.choice, Quot.sound]
FrohmanianTether.RegularField.cont	[propext, Classical.choice, Quot.sound]
FrohmanianTether.RegularField.diff	[propext, Classical.choice, Quot.sound]
FrohmanianTether.RegularField.div_free	[propext, Classical.choice, Quot.sound]
FrohmanianTether.RegularField.memL2	[propext, Classical.choice, Quot.sound]
FrohmanianTether.kineticEnergyCurlIdentityReg_of_regularStream	[propext, Classical.choice, Quot.sound]
FrohmanianTether.pairing_integrable	[propext, Classical.choice, Quot.sound]
FrohmanianTether.regularField_sub	[propext, Classical.choice, Quot.sound]
FrohmanianTether.regularFunctionalDerivative_eq	[propext, Classical.choice, Quot.sound]
FrohmanianTether.regularRep_unique	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_of_l2_energy_zero	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.smul_coord	[propext, Classical.choice, Quot.sound]
ArnoldGeometric.CoadjointOrbit.mk.congr_simp	[propext, Classical.choice, Quot.sound]
R3BiotSavart.biotSavartKernel_eq_profile	[propext, Classical.choice, Quot.sound]
R3BiotSavart.biotSavartKernel_of_ne'	[propext, Classical.choice, Quot.sound]
R3BiotSavart.cross_add_smul	[propext, Classical.choice, Quot.sound]
R3BiotSavart.curl_zero	[propext, Classical.choice, Quot.sound]
R3BiotSavart.exists_const_not_integrable	[propext, Classical.choice, Quot.sound]
R3BiotSavart.norm_le_sum_norm_cross	[propext, Classical.choice, Quot.sound]
R3BiotSavart.norm_sq_cross_single	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_integrable_norm_rpow_neg_two	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_isGateauxRepresentative_kineticEnergy	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_kineticEnergyCurlIdentityReg	[propext, Classical.choice, Quot.sound]
R3BiotSavart.not_regularStreamRepresentative	[propext, Classical.choice, Quot.sound]
R3BiotSavart.orbit_zero_smul	[propext, Classical.choice, Quot.sound]

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